Stabilization of Black Cotton Soil with Alkali Bypass dust

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Abstract

Black Cotton soil is problematic since it exhibits an alternate shrink and swell behavior. After conducting a literature survey of the existing stabilization methods of Black Cotton soils, a gap was identified when using Alkali Bypass dust as a potential stabilizing material. Alkali Bypass dust is a by-product of the soda ash industry, which is explored to improve geotechnical properties like the strength and plasticity of Black Cotton soil. This study conducts laboratory tests like - the Modified Proctor Compaction test, Atterberg's limits test, and California Bearing Ratio (CBR) test to determine the change in characteristics of the mix. The results indicate a decrease in the maximum dry density from 1.65 g/cc to 1.44 g/cc and a reduction in the optimum moisture content and plasticity index from 19.5% to 23.5% and 34% to non-plastic, respectively, as the proportion of Alkali Bypass dust increases from 0% to 50% of the mix. The CBR value shows a maximum value for the 70:30 combination, i.e., 42%..

Introduction

- Black Cotton soil exhibits expansive soil properties due to clay minerals, thus, severe damage is seen in foundations and structures built over it.
- Alkali Bypass dust is a by-product produced by the soda ash manufacturing plant.
- Laboratory tests were conducted on the samples with a proportion of Alkali Bypass dust as 0%, 10%, 30%, and 50% of the total mixture.
- Initially, the Grain Size distribution of the Black Cotton soil and the Alkali Bypass dust was carried out.
- The Modified Proctor compaction test was conducted to determine the Optimum Moisture Content (OMC) and Maximum Dry Density (MDD) of the mix. Atterberg's limits tests were performed to determine the Liquid and Plastic limits of the mixtures. CBR sample of Black Cotton soil was tested after 4 days of soaking in water, and the different mixes were tested after 7 days of curing under moist sand followed by 4 days of immersion in water.

Literature Survey

Authors	Paper Title	Stabilizer Used	Results
Akshatha, R., and H. M. Bharath (2016)	Improvement in CBR of Black Cotton Soil Using Brick Powder (Demolition Brick Masonry Waste) and Lime	Demolition Brick Masonry Waste	OMC increases, MDD decreases and CBR value of the mix increases on addition of Brick Powder and lime to the soil
Al-Khafaji, Ruqayah, Anmar Dulaimi, Hassnen Jafer, Nuha S. Mashaan, Shaker Qaidi, Zahraa Salam Obaid, and Zahraa Jwaida (2023)	Stabilization of Soft Soil by a Sustainable Binder Comprises Ground Granulated Blast Slag (GGBS) and Cement Kiln Dust (CKD)	Ground Granulated Blast Furnace Slag (GGBS) and Cement Kiln Dust (CKD)	The optimum binder combination was found out to be 6% binder (75% CKD and 25% GGBS) which gave maximum increase in strength
Agapitus Ahamefule Amadi (2014)	Enhancing durability of quarry fines modified black cotton soil subgrade with cement kiln dust stabilization	Cement Kiln Dust (CKD)	CKD reduced the moisture susceptibility of the overall mix and the CBR value increased for CKD of 8% to 16%
Chansoria, Aditya, et al (2016)	Effect of quarry dust on engineering properties of black cotton soil.	Quarry Dust	OMC value decreased, MDD value increased and the CBR value increased for the mix
Naphol Yoobanpot, Pitthaya Jamsawang, Suksun Horpibulsuk (2017)	Strength behavior and microstructural characteristics of soft clay stabilized with cement kiln dust and fly ash residue	Cement Kiln Dust (CKD) and Fly Ash (FA)	The Optimum composition for strength development was determined as 13% CKD with partial replacement of 20% FA

Results and Discussions

Table.1 Geotechnical characteristics of BC soil, and BC soil mixed with Alkali Bypass dust

Mix Composition (BC Soil : Alkali Bypass dust)	Optimum Moisture Content (%)	Maximum Dry Density (g/cc)	Liquid Limit (%)	Plasticity Index (%)	CBR Value (%)
100:0	19.5	1.65	69.0	34.0	2.5
90:10	21.0	1.60	62.0	28.0	11.0
70:30	24.0	1.57	48.0	12.0	42.0
50:50	23.5	1.44	48.0	Non-Plastic	38.0
0:100	25.0	1.38	45.0	Non-Plastic	-

- The geotechnical properties were determined as per to IS 2720.
- From the Grain Size analysis of the Black Cotton soil the percentage of sand, silt and clay was found to be 5%, 57%, 38% respectively.
- The Alkali Bypass dust was available in powder form. The liquid limit of the mix is indicated in Table.1 and Fig.1.
- As per IS Classification, Black Cotton soil may be classified as Inorganic clay having high compressibility, and Alkali Bypass dust indicates non-plastic material.
- The addition of Alkali Bypass Dust to the Black Cotton soil shows an increase in the OMC and a decrease in MDD.
- This is a result of the fact that the Alkali Bypass Dust is a low-density powder, which is why the addition of it to the mix causes a decrease in the overall density of the mix.
- The CBR value increased significantly after adding Alkali Bypass dust to the Black Cotton soil, as shown in Table.1.
- The workability of the Black Cotton soil increases with the addition of the Alkali Bypass dust, and the mixture goes from being highly plastic to Non–Plastic.

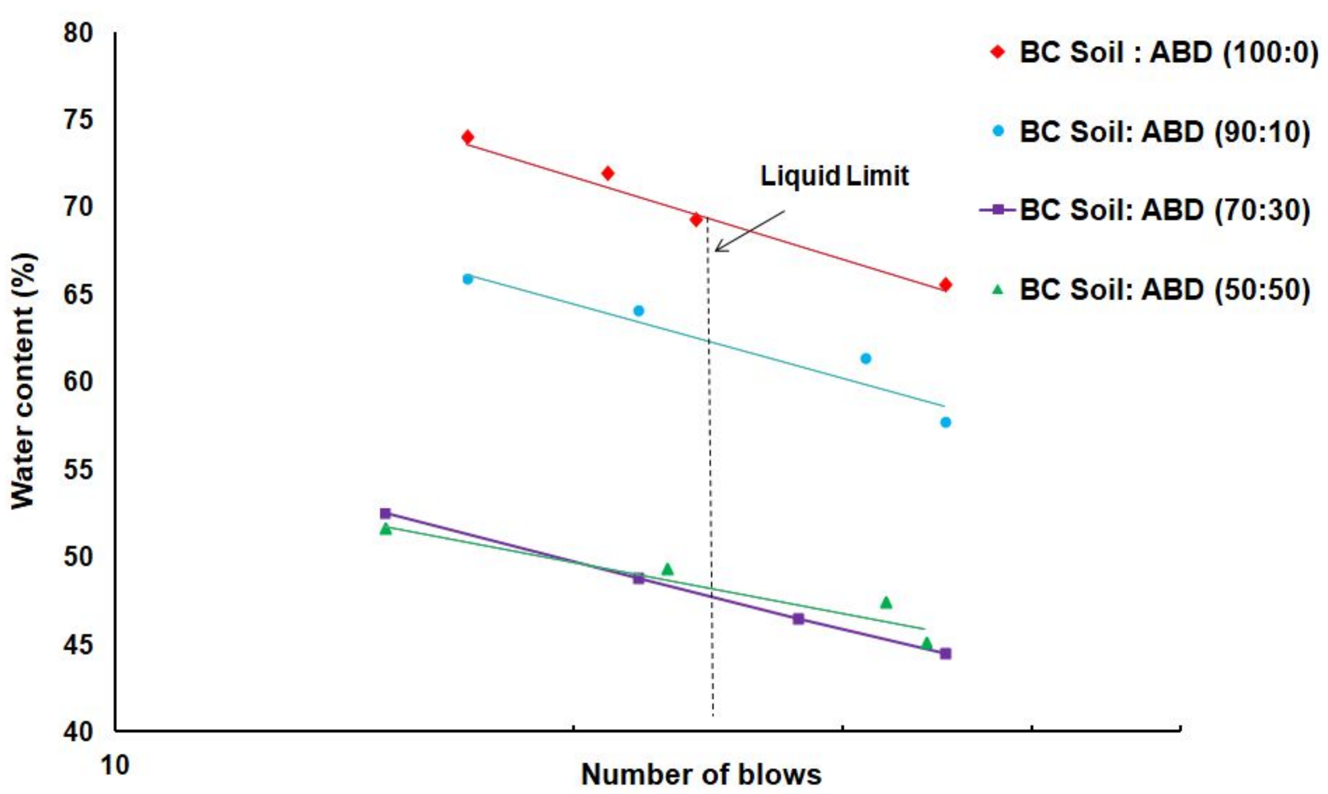

Fig. 1 Flow curve to determine the liquid limit for different mixes

Conclusion and Future Scope

- The results obtained so far indicate an improvement in the plasticity characteristics of the Black Cotton soil using Alkali Bypass dust.
- This is shown by a steady decrease in the Plasticity Index by increasing the proportion of Alkali Bypass dust in the mixture, thus making it more workable.
- The Alkali Bypass dust consists primarily of non-plastic silt-sized particles with low MDD and high OMC. It shows self-hardening properties when it comes in contact with water.
- The CBR value shows a peak at the 70:30 composition and then decreases slightly with the addition of Alkali Bypass dust.
- Thus, some strength is compromised due to the decrease in density of the entire mix by the addition of this low density powder.
- Thus, Alkali Bypass dust can be used for subgrade stabilization and embankment fill construction due to its lightweight and non-plastic nature.

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References

1. Agapitus Ahamefule Amadi, Enhancing durability of quarry fines modified black cotton soil subgrade with cement kiln dust stabilization, Transportation Geotechnics, Volume 1, Issue 1,2014, Pages 55-61, ISSN 2214-3912.
2. Akshatha, R., and H. M. Bharath. "Improvement in CBR of Black Cotton Soil Using Brick Powder (Demolition Brick Masonry Waste) and Lime." *International Journal of Innovative Research in Science, Engineering and Technology* 5, no. 9 (2016).
3. Al-Khafaji, Ruqayah, Anmar Dulaimi, Hassnen Jafer, Nuha S. Mashaan, Shaker Qaidi, Zahraa Salam Obaid, and Zahraa Jwaida. 2023. "Stabilization of Soft Soil by a Sustainable Binder Comprises Ground Granulated Blast Slag (GGBS) and Cement Kiln Dust (CKD)" *Recycling* 8, no. 1: 10.
4. Chansoria, Aditya, et al. "Effect of quarry dust on engineering properties of black cotton soil." *Int. J. Innov. Res. Sci. Technol* 2.11 (2016): 715-718.ISSN (online): 2349-6010
5. IS : 2720 (Part 4)- 1985: Indian Standard. METHODS OF TEST FOR SOILS. PART 4- GRAIN SIZE ANALYSIS. (Second Revision).
6. IS:2720 (Part 5)-1985: Indian Standard. METHOD OF TEST FOR SOILS. PART 5- DETERMINATION OF LIQUID AND PLASTIC LIMIT. (Second Revision)
7. IS 2720 (Part 8)- 1983: Indian Standard. METHODS OF TEST FOR SOILS. PART 8- DETERMINATION OF WATER CONTENT-DRY DENSITY RELATION USING HEAVY COMPACTION.
8. IS: 2720 (Part 16)-1973: Indian Standard. METHODS OF TEST FOR SOILS. PART 16- LABORATORY DETERMINATION OF CBR.
9. Naphol Yoobanpot, Pitthaya Jamsawang, Suksun Horpibulsuk, Strength behavior and microstructural characteristics of soft clay stabilized with cement kiln dust and fly ash residue, Applied Clay Science, Volume 141,2017, Pages 146-156, ISSN 0169-1317.